Towards quantisation of Boussinesq solitons: Phase 2 master report (Routes A–E, Phases 2a–2h)

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Abstract

Following the PI's initial decision, we first studied the *bad* Boussinesq equation

$$v_{tt} = v_{xx} + v_{xxxx} + 3(v^2)_{xx},$$

whose soliton sector is regular and purely elastic, has regular breathers, and is the continuum limit of the quantum Toda chain (Phase 1), and subsequently extended the analysis to the *good* (well-posed) Boussinesq equation $u_{tt} = u_{xx} - u_{xxxx} - 3(u^2)_{xx}$ and its Corrigan–Zambon integrable defects (Phases 2c–2h).

Semiclassical ($O(\hbar^{-1})$) results (Route A). Two-soliton scattering is an exactly symplectic map for the canonical structure of the field theory, *only* once ordering-dependent "topological" cross terms $d\Pi_i \wedge dM_j$ are included. These terms come from the kink in the potential ϕ. With them we obtain a closed-form semiclassical two-soliton phase, $S \simeq e^{i\delta/\hbar}$:

$$\delta(1,2) = \tfrac{2}{9}(\ell_1 - \ell_2)\log A_{12} + \tfrac{2}{3}M_1 M_2(v_1 - v_2), \qquad \ell = \epsilon\cosh 3\beta \text{ (bad)}, \quad \ell = -\epsilon\cos 3\alpha \text{ (good)}.$$

- δ is antisymmetric (unitarity) and obeys a crossing-type relation under $\beta \to \beta + i\pi$.
- Continued to complex conjugate rapidities, $i\cdot\frac{2}{9}(\ell_1 - \ell_2)$ reproduces *exactly* the adiabatic invariant of the Boussinesq breather computed independently by Bogdan and Charkina [1], turning breather quantisation into a string condition $\ell_1 - \ell_2 = -\frac{9i}{2}N\hbar$.

One-loop, Toda, and defect results after independent audits. The continuum phase integrals have closed dilogarithmic expressions, and the regulated co-moving mode sum supports the proposed one-loop charge Lagrangian when its cutoff endpoint is retained. These are checked results **(c)**; the complete projected fluctuation determinant and absolute scattering-phase normalisation remain unestablished **(d)**. The quantum Toda calculations supply a separate small-amplitude test, but a common quantum momentum/frame dictionary and an independent derivation of the proposed leading residual are still required **(d)**. The KP chord identities are algebraic results **(c)**. Their identification with physical phases in every orientation, and their use as quantum defect fusion rules, require additional boundary, state-normalisation, and stability input **(d)**. In particular, neither exact one-loop reconciliation nor an all-loop S-matrix is claimed here.

Audit status. This revision incorporates `independent_phase2h_audit.pdf`, `audit_charges_action.pdf`, `audit_determinant_foundations.pdf`, `audit_frames_defects.pdf`, and the follow-up notes `audit_projected_measure.pdf` and `audit_toda_dictionary.pdf`. Their independent scripts, precision, failed claims, and limitations are recorded in those notes. The historical phase write-ups remain available; the qualifications in this report supersede their stronger physical conclusions. No new regularisation or phase convention is selected.

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1 Setting

Throughout, $v = 2(\log\tau)_{xx}$ and $k = \sinh\beta$. A soliton has velocity $v = \epsilon\cosh\beta$, with $\epsilon = +1$ for right-movers and $\epsilon = -1$ for left-movers. The potential is $v = \phi_x$ and the Lagrangian is

$$\mathcal{L} = \tfrac{1}{2}\phi_t^2 - \tfrac{1}{2}\phi_x^2 + \tfrac{1}{2}\phi_{xx}^2 + \phi_x^3,$$

with $\pi = \phi_t$ and canonical symplectic form $\Omega = \int \delta\pi \wedge \delta\phi$.

On one soliton the charges are:

- mass $M = \int v = 2k$;
- ϕ-shift charge $\Pi = \int \pi = -vM$, conserved because $\phi \to \phi + c$ is a symmetry;
- momentum $P = -\int \pi\phi_x = \frac{2}{3}\epsilon k^3\cosh\beta$;
- energy $E = \frac{2}{15}k^3(5 + 4k^2)$.

These satisfy $dE/dP = v$. **(a)** [1]/**(c)**

The two-soliton constants are

$$A_{\mathrm{ov}} = \frac{1 + 3\coth^2\frac{\beta_1+\beta_2}{2}}{1 + 3\coth^2\frac{\beta_1-\beta_2}{2}}, \qquad A_{\mathrm{ho}} = \frac{1 + 3\tanh^2\frac{\beta_1-\beta_2}{2}}{1 + 3\tanh^2\frac{\beta_1+\beta_2}{2}},$$

both in $(0, 1)$, and the position shifts are $\Delta x_1 = -k_1^{-1}\log A$, $\Delta x_2 = +k_2^{-1}\log A$ (soliton 1 starts on the left). **(c)** (Phase 1)

What a soliton S-matrix would mean here. The model has no boost symmetry and is gapless (phonons with $\omega \simeq |k|$). The natural object is therefore the non-relativistic two-body S-matrix $S(P_1, P_2)$ of soliton asymptotic states, in the sense of Faddeev–Korepin [7]. Classical scattering is purely transmissive and diagonal, so S is a phase; for two species (left and right movers) S is diagonal.

For diagonal scattering the Yang–Baxter equation is automatic. The non-trivial requirements are:

- unitarity;
- the correct semiclassical limit;
- analyticity and a crossing substitute;
- a bound-state (breather) spectrum consistent with poles.

2 Route A: semiclassical quantisation

2.1 The symplectic structure of soliton scattering

Proposition 2.1 (**(c)**). *Consider two well-separated solitons, with ϕ normalised by $\phi(-\infty) = c$. Up to exponentially small terms,*

$$\Omega = d\Pi_{\mathrm{tot}} \wedge dc + \sum_j dP_j \wedge dX_j + \big[X_i > X_j\big]\, d\Pi_i \wedge dM_j.$$

The argument is as follows. π_i is localised at X_i, and there ϕ_j equals 0 or M_j according to whether soliton i is to the left or right of soliton j. The self-term $dP \wedge dX$ was derived in Phase 1. In sine-Gordon the analogous cross term vanishes, because the kink height $2\pi/\beta$ is fixed. Here $M = 2k$ varies with the soliton parameter.

Proposition 2.2 (**(c)**, `scripts/symplectic_scattering.py`)**.** *The classical scattering map $X_j \mapsto X_j + \Delta x_j$ (with k_j and c fixed) preserves Ω, i.e.*

$$\sum_j dP_j \wedge d\Delta x_j = (\textit{cross term})_{\text{in}} - (\textit{cross term})_{\text{out}}.$$

This was verified to 40 digits for overtaking and head-on collisions at five parameter points each. It was also verified for the good *equation in its regular region (`scripts/consistency_checks.py`). Without the cross terms the identity fails by $O(1)$ amounts; for example, the left side equals* 19.79 *at $\beta = (0.9, 0.3)$.*

This is a non-trivial confirmation of the classical picture. It also shows that a semiclassical S-matrix built from the naive rule "$\Delta x_j = -\partial\delta/\partial P_j$" does not exist: the shifts are not curl-free in the canonical momenta.

2.2 The semiclassical phase

In each ordering region we choose Darboux coordinates by the rule

$$X_j' = X_j + \gamma_j \sum_{i \text{ left of } j} M_i, \qquad \gamma_j = \frac{d\Pi_j/dk_j}{dP_j/dk_j}.$$

The mirrored rule (normalising ϕ at $+\infty$) gives the same in–out differences. The shifts $\Delta x_j'$ are then curl-free. Their potential is the semiclassical phase δ, with $\Delta x_j' = -\partial\delta/\partial P_j$ and $S \simeq \exp(i\delta/\hbar)$, by stationary phase on two-particle wave packets [7].

Proposition 2.3 (**(c)**, checked semiclassical phase)**.** *1.* $\dfrac{dP}{d\beta} = \frac{2}{3}\,\epsilon k \sinh 3\beta$ *for the bad equation. For the good equation,* $\dfrac{dP}{d\alpha} = \frac{2}{3}\,\epsilon k \sin 3\alpha$ *with $k = \sin\alpha$; this vanishes at the cusp $\alpha = \pi/3$ found in Phase 1.*

2. For both overtaking and head-on collisions, and both equations,

$$\boxed{\delta(1,2) = \tfrac{2}{9}\,(\ell_1 - \ell_2)\,\log A_{12} + \tfrac{2}{3}\,M_1 M_2\,(v_1 - v_2)} \tag{1}$$

with $\ell = \epsilon\cosh 3\beta$ (bad) or $\ell = -\epsilon\cos 3\alpha$ (good). The remainder was first found by symbolic integration, then identified as $\frac{2}{3}M_1M_2(v_1 - v_2) = -\frac{2}{3}(\Pi_1 M_2 - \Pi_2 M_1)$ by exact matching.

3. Unitarity: $\delta(1,2) = -\delta(2,1)$.

4. Crossing-type relation: $\delta_{\text{ov}}(\beta_1, \beta_2 + i\pi) = -\delta_{\text{ho}}(\beta_1, \beta_2)$. *The shift $\beta \to \beta + i\pi$ maps a right-mover to a left-mover with $M \to -M$.*

Checks: `scripts/semiclassical_phase.py` and `scripts/consistency_checks.py`.

Remark 2.4.
- ℓ is proportional to the spectral parameter of the third-order Lax operator. In the uniformisation of [4], $\lambda = \frac{1}{2}(k^3 + k^{-3})$, with k^3 playing the role of $e^{3\beta}$.
- The dynamical part of δ is (difference of spectral parameters)$\times \log A$.
- The second part is a "mutual statistics" phase: an Aharonov–Bohm-like phase $\propto \Pi_1 M_2 - \Pi_2 M_1$ between objects carrying ϕ-shift charge and ϕ-winding. **(d)** (interpretation)

Remark 2.5 (KdV limit, **(c)**). For small amplitudes the topological term is of the *same order* as the KdV-type term $2\beta_1 \log A$ in $\partial\delta/\partial\beta_1$. Numerically, their ratio tends to 0.564 for $\beta_1/\beta_2 = 1.7$. The heuristic of Bonnemain and Doyon [2] identifies the quantum phase derivative with $P'_{\text{GHD}}\Delta$, with $P_{\text{GHD}} = \eta^2/2$. That identification reproduces our dynamical term but *misses the topological term.* The classical generalised hydrodynamics, which uses only Δx, is unaffected. **(c)**

2.3 Bohr–Sommerfeld: breathers as spectral-parameter strings

Bogdan and Charkina quantise the Tajiri–Murakami breather of the bad equation by Bohr–Sommerfeld, $I_{\mathrm{br}} = N\hbar$, with an independently computed adiabatic invariant I_{br} **(a)** [1, (64)–(65)].

Proposition 2.6 (**(c)**). *Set $\beta_{1,2} = a \pm ib$, which is Bogdan–Charkina's parametrisation $\kappa \pm ik = \sinh(a \pm ib)$. Then:*

1. *A_{12} equals their a_{12} [1, (33)]. Regularity $A > 1$ holds if and only if $b > \pi/3$, which is their existence boundary.*
2. *$i \cdot \frac{2}{9}(\ell_1 - \ell_2) = I_{\mathrm{br}}$ exactly, with $I_{\mathrm{br}} = -\frac{4}{9}\sinh 3a \sin 3b$, in agreement with [1, (64)].*

Hence the coefficient of $\log A$ in (1) is the breather action. Bohr–Sommerfeld quantisation reads

$$\ell_1 - \ell_2 = -\tfrac{9i}{2}\, N\hbar, \qquad N \in \mathbb{N}, \tag{2}$$

which is a string condition in the spectral parameter ℓ.

This is the strongest cross-check in this report. It ties our two-soliton phase, built from Hirota shifts and the canonical symplectic form, to an independent quasiclassical calculation in the literature. Item 2 was checked at three parameter points to 10^{-16}.

Equation (2) is the form in which bound states appear in Bethe-ansatz solvable models. It suggests that ℓ is the variable in which an exact S-matrix has poles: $\ell_1 - \ell_2 \in -\frac{9i}{2}\hbar\mathbb{N}$, possibly with $\hbar$ renormalised. **(d)**

2.4 What Route A gives and subsequent updates (Phases 2c, 2e, 2h)

- *Obtained at tree level ($O(\hbar^{-1})$).* The leading semiclassical phase (1), which is unitary, crossing-type covariant and consistent with breather quantisation. Also an asymptotic Bethe-ansatz quantisation of N solitons on a ring, $P_j\Lambda + \sum_{k\neq j}\delta(j,k) = 2\pi\hbar n_j$. **(d)**: the twisted boundary conditions forced by ϕ-winding on a ring have not been worked out. Unlike in the relativistic Toda case [12, 13], the tree-level phase δ is *elementary*: no dilogarithm appears at $O(\hbar^{-1})$, because $dP \propto k\sinh 3\beta\, d\beta$ and integration by parts closes.

- *Update (Phase 2c, `smatrix_ansatz.pdf`): Candidate Gamma ansatz.* Setting $\eta = \frac{2}{9}(\ell_1 - \ell_2)/\hbar$ and $R(x) = \Gamma(1+ix)/\Gamma(1-ix)$, a candidate exact S-matrix was proposed **(d)**: $S = R(\eta)(e^2A/\eta^2)^{i\eta}e^{iT/\hbar}$ (overtaking) and $S = R(\eta)^{-1}(A\eta^2/e^2)^{i\eta}e^{iT/\hbar}$ (head-on). Its semiclassical expansion gives $\arg S - \delta/\hbar \to \pm\frac{\pi}{2}\operatorname{sgn}\eta + O(\hbar)$ and its poles match the Bogdan–Charkina breather levels (2).

- *Update (Phase 2d, `oneloop.pdf`): Bad-equation one-loop DHN analysis.* For the bad equation, modes with $|q| > 1$ are exponentially unstable ($|A(\mathrm{sol},\mathrm{ph})| \neq 1$), so the DHN sum is well posed only on the Charlier–Lenells invariant stable subspace $|q| < 1$ (Scheme A). There the breather one-loop energy and momentum are holomorphic continuations of the constituent soliton values, so the string condition (2) receives no $O(\hbar^0)$ shift at fixed rapidity; however, a reversed-group-velocity mode at $q^* = -\sqrt{3}/2$ is captured by slow solitons, making Scheme A differ from the Toda continuum limit at $\beta \to 0$. **(c)**

- *Update (Phase 2e, `oneloop_phase.pdf`): Continuum one-loop integral.* For the good equation, the tau-function construction gives reflectionless phonon modes and a three-root transmission phase. The corresponding Faddeev–Korepin continuum integral evaluates to a sum of Legendre χ_2 and logarithmic terms **(c)**. Independent potential-field quadrature confirms the spectral-density identity; it does not establish completeness, the collective-coordinate measure, or the full time-dependent determinant (`audit_`

`determinant_foundations.pdf`). The nonconstant integral is therefore a conditional test of the continued Gamma and pure-exponential candidates, not a completed physical exclusion **(d)**. The separate projected-measure audit now proves factorisation of the finite collective symplectic density despite the tail cross terms, and verifies its Wronskian matrix on the exact collision background **(c)**(`audit_projected_measure.pdf`). Its explicit projector does not preserve a fixed-potential-endpoint fluctuation domain **(c)**. The field and constraint measures and the time-boundary determinant therefore remain open **(d)**.

- *Update (Phase 2h, `charge_frame_master.pdf`): One-loop soliton charge audit.* Because the Boussinesq soliton profile $u_0(x-vt;\alpha)$ changes width $k=\sin\alpha=\sqrt{1-v^2}$ with velocity $v=\epsilon\cos\alpha$, the fixed-profile mode-sum charges $(\Delta E_1^{\mathrm{ren}},\Delta P_{\mathrm{mode}})$ have a non-zero velocity anomaly $\mathcal{A}_v(\alpha)=d\Delta E_1^{\mathrm{ren}}/d\alpha - v\,d\Delta P_{\mathrm{mode}}/d\alpha \neq 0$. In the co-moving one-loop effective Lagrangian $L_1(v)=v\Delta P_{\mathrm{mode}}-\Delta E_1^{\mathrm{ren}}$, the total weight $\cos(2\phi_r)-\cos(\alpha+\phi_r)$ vanishes identically on the light root $\phi_3=\alpha$, cancelling all $\ln\alpha$ singularities to all orders and giving the following conditional collective-coordinate momentum **(c)**

$$\Delta P_{\mathrm{can}}(\alpha,\epsilon)=\frac{\partial L_1}{\partial v}=-\epsilon\frac{\sqrt{3}\cos\alpha}{\pi}\ln\left|\frac{\tan\frac{\alpha+2\pi/3}{2}}{\tan\frac{\alpha-2\pi/3}{2}}\right|=-\frac{4\epsilon}{\pi}\alpha+\frac{8\epsilon}{9\pi}\alpha^3-\frac{8\epsilon}{45\pi}\alpha^5+O(\alpha^7),$$

which satisfies $d\Delta E_{\mathrm{can}}/dv = v\,d\Delta P_{\mathrm{can}}/dv$ identically. The regulated mode sum reproduces L_1 when its cutoff endpoint is included (`audit_charges_action.pdf`, **(c)**). Identifying its derivative with the renormalised field-theory momentum still requires the projected determinant, stationary-profile correction, and boundary terms **(d)**.

3 Route B: quantum inverse scattering on the line

What would have to be true. The Lax matrix $U(x,\lambda)$ would need an ultralocal r-matrix Poisson algebra, so that it can be regularised on a lattice and quantised à la Faddeev–Korepin–Sklyanin. Then a Bethe ansatz over a pseudo-vacuum with soliton excitations would give S.

Proposition 3.1 (**(c)**, `scripts/routes_BC_checks.py`)**.** *In canonical variables the Lax data of Phase 1 are* $q=\frac{3}{2}\phi_x-\frac{1}{4}$ *and* $r=\frac{1}{2}q_x+\frac{\sqrt{3}}{4}\pi$ *(good-equation real form; the bad one differs by* i*'s). Their brackets are*

$$\{q(x),r(y)\}=\tfrac{3\sqrt{3}}{8}\,\delta'(x-y),\qquad \{q,q\}=\{r,r\}=0.$$

So the companion-matrix Lax connection is non-ultralocal.

Where it stalls. There are two known ways out, and neither yet gives a theory on the line.

1. Maillet's r/s formalism with lattice regularisation [14]. No construction exists for Boussinesq.
2. The Miura (free-field) map. This is the route of Bazhanov–Hibberd–Khoroshkin [3] and Hikami [10]. It yields $U_q(\widehat{\mathfrak{sl}}_3)$ T- and Q-operators, but in the chiral conformal setting on a circle, with no particle sector.

Dimakis et al.'s Yang–Baxter maps [6] concern polarisations in the matrix case and are vacuous for the scalar equation. The "beyond Yang–Baxter" remark there is therefore not an obstruction here. For diagonal scattering, Yang–Baxter holds trivially. **(c)**

4 Route C: W_3, sl_3 affine Toda and the $U_q(\widehat{\mathfrak{sl}}_3)$ R-matrix

What would have to be true. Boussinesq solitons would have to be identifiable as a limit of $a_2^{(1)}$ (complex) affine Toda solitons, whose S-matrices are the $U_q(\widehat{\mathfrak{sl}}_3)$ intertwiners of Hollowood [12] and Gandenberger [8].

Proposition 4.1 (**(c)**, `scripts/routes_BC_checks.py`)**.** *Take Hollowood's $a_2^{(1)}$ interaction constant [11, (1.19)], $A^{(12)}(\theta) = \frac{2\cosh\theta+1}{2(1+\cosh\theta)}$, between the two fundamental species. With $S = \beta_1+\beta_2$ and $D = \beta_1 - \beta_2$,*

$$A_{\mathrm{ov}}^{\mathrm{bad}}(\beta_1,\beta_2) = \frac{A^{(12)}(S)}{A^{(12)}(D)} \cdot \frac{\tanh^2(D/2)}{\tanh^2(S/2)}, \qquad A_{\mathrm{ho}}^{\mathrm{bad}}(\beta_1,\beta_2) = \frac{1}{A_{\mathrm{ov}}^{\mathrm{bad}}(\beta_1,\beta_2+i\pi)}.$$

Here $\tanh^2(\theta/2)$ is the sine-Gordon soliton–soliton constant. Equivalently,

$$A_{\mathrm{ov}} = \frac{G(S/2)}{G(D/2)}, \qquad G(y) = \frac{16\prod_{a=1,2}\sinh\frac{y+ia\pi/3}{2}\sinh\frac{y-ia\pi/3}{2}}{\sinh^2 y}.$$

So the Boussinesq interaction is built from exactly the $\mathfrak{sl}_3$ blocks, but in the form "$F(\beta_1 - \beta_2)/F(\beta_1+\beta_2)$". This is the structure of scattering off a particle *and its mirror image*. It suggests that a Boussinesq S-matrix might take the form

$$S(\beta_1,\beta_2) = \mathcal{S}(\beta_1-\beta_2)\,\mathcal{S}(\beta_1+\beta_2+i\pi)^{\pm 1}$$

with $\mathcal{S}$ built from $U_q(\widehat{\mathfrak{sl}}_3)$ and sine-Gordon factors. Such a form is reminiscent of boundary (reflection) amplitudes. That would connect to the "boundary affine Toda" structure in Bazhanov–Hibberd–Khoroshkin [3]. **(d)**

The obstruction is that the semiclassical phase (1) is not of this product form. Its weight $\frac{2}{9}(\ell_1-\ell_2)$ is not a sum of functions of S and D, and its topological part has no Toda analogue.

Further obstructions:

- Toda solitons are relativistic with fixed masses m_a. In the Toda constant the mass term $4m^2\sin^2(\pi a/3)$ appears where Boussinesq has K^4, so the species label is "replaced" by the wavenumber. There is no limit of fixed-species Toda solitons that yields Boussinesq solitons. **(c)**
- Takács and Watts show that $a_2^{(1)}$ soliton–breather scattering is flawed both classically and quantum mechanically **(b)** [20].

Route C stalls: the structural similarity is precise, but no dictionary is available. What is lost in the non-relativistic reduction is the difference form, and with it the standard crossing.

5 Route D: discrete models and the quantum Toda Fermi-edge limit

(i) Quantum discrete Boussinesq (Hikami [10]). This is the Belov–Chaltikian lattice W_3 chain, periodic, with a Baxter Q-operator. A soliton/particle sector would require a thermodynamic limit with an identified ground state and excitations. That has not been done in the literature, and we did not attempt it. *Stalls immediately*; it is a research project in itself.

(ii) Quantum Toda chain. This is our recommended route for the small-amplitude corner. The required ingredients are:

1. The exact quantum Toda chain **(a)** [9, 17] and its asymptotic Bethe ansatz **(a)**/**(b)** [19, 15, 18]. Its particle excitations become solitons in the classical limit **(b)** [15, 16].

2. The dressed particle–particle S-matrix in the finite-density ground state.

3. Its classical limit should reproduce the Toda soliton phase.

4. The small-amplitude scaling limit should give (1).

Update (Phase 2b, `toda_route.pdf`). Steps 2–4 were carried out in the classical limit, and step 2 at finite $\hbar$ to $O(\hbar)$. **(c)**

- The classical limit of the Bethe ansatz reproduces the Toda soliton energies, momenta and exact two-soliton shifts.
- Its two-body phases carry exactly the predicted topological cross terms.
- The label-frame phase is $\delta_T = (p_1 - p_2)\log A_T - (\Pi_1 M_2 - \Pi_2 M_1)$.
- In the scaling limit, $\delta_T \to 3b\,\delta_{\mathrm{bad}}$ for overtaking *and* head-on collisions. This independently confirms (1), including its topological term, from an exactly solvable quantum model.

Update (Phases 2d, 2f, 2h): Fermi-edge limit and open quantum matching. The existing numerical implementation fixes $y_j = \beta_j^2/\hbar_B = b\kappa_j^2/\hbar_T$ as $\hbar_T = 3b\hbar_B \to 0$, and uses the classical frame conversion (`scripts/toda_edge_ba.py`, `scripts/matching_toda.py`). **(c)**

- *Head-on channel:* the reported small residual is evidence within that implemented dictionary. The vanishing small-amplitude continuum integral alone does not prove matching at fixed physical quantum momenta **(d)**(`matching.pdf`).
- *Overtaking channel:* the numerical comparison in `toda_edge.pdf` finds a discrepancy from the Gamma candidate within the chosen dictionary **(c)**. Phase 2h proposes, for $y_1 = Y$, $y_2 = r^2 Y$,

$$c_0^*(r) = -\frac{4}{\pi}\big(\chi_2(r) - r\big), \qquad \lim_{Y\to\infty} R_0(Y, r) \stackrel{?}{=} c_0^*(r). \tag{3}$$

  The limiting equality is a conjecture **(d)**. Agreement with central fitted intercepts is smaller than the spread among saved edge-extrapolation alternatives; it is not an analytical derivation. The script's purported intrinsic loop contribution is defined using this target, so cannot independently establish it (`independent_phase2h_audit.pdf`, **(c)**).
- *Frame and remainder checks:* the classical frame limit is verified, but its quantum correction is not derived **(d)**. At fixed ground state and y_1, the implemented edge reference is independent of r; it cannot alone generate the $4r/\pi$ term attributed to it **(c)**(`audit_frames_defects.pdf`). Conditional on the proposed logarithmic momentum shift, the classical second-order Taylor term contains $(\log Y)^2/Y$ **(c)**. Cancellation in the full phase remains to be shown; the previously quoted pure $1/Y$ coefficient is not resolved by the saved fits.

Follow-up: an exact quantum endpoint diagnostic. The independent Toda dictionary audit derives $P_{\mathrm{dr}}(q) = \pi\hbar_T/a$ and an endpoint bilinear for the symmetric part of the directional dressed phase from the existing Fredholm equations **(c)**(`audit_toda_dictionary.pdf`). The edge spectral coordinate is not the dressed physical momentum; their Jacobian must be retained. A converged finite-$\hbar_T$ mixed-difference check finds a nonseparable symmetric remainder after the implemented classical frame subtraction, reproduced with the original solver **(c)**. At the fixed momentum dictionary, one-body phase-reference terms cannot remove that remainder. This is a reciprocity diagnostic, not an assertion about a second physical collision or a proof that the correction survives the nested edge limit. The antisymmetric quantum frame contribution and its identification with continuum charges remain unestablished **(d)**.

6 Route E: integrable defects and KP chord identities

Corrigan and Zambon (CZ) constructed integrable type-I Bäcklund defects for the Boussinesq equation **(a)** [5]. Each defect carries a single KP root $q_D = -s/(2\sqrt{3})$ ($s = \eta\sigma$, $\eta = \pm 1$), whereas a soliton carries a KP chord (p_a, p_b) connecting two roots of the KP cubic $4q^3 - q - \ell/(3\sqrt{3}) = 0$.

Semiclassical soliton–defect transmission (Phase 2c). (c)

- The classical soliton–defect transmission phase is $\delta_D = \frac{2}{9}(\ell - \ell_D)\ln|z_t| - \frac{2}{3}(\Pi M_D - \Pi_D M)$ with $z_t = ((p_b - q_D)/(p_a - q_D))^{\epsilon\eta}$ and $\ell_D = (3s - s^3)/2$. The static defect adds no ordering-dependent cross terms; two-soliton + defect scattering is symplectic and factorises.
- A defect vacuum q_D is linearly stable if and only if $s = \eta\sigma < -1$ ($q_D > 1/(2\sqrt{3})$), which selects the unique ground state in each KP triple $\{q_1, q_2, q_3\}$.
- Along any edge $q_i \to q_{i+1}$ of a KP triangle where a soliton S_i on $\{q_i, q_{i+1}\}$ is absorbed by the defect, the tree-level phases satisfy exact defect fusion $\delta_D(q_{i+1})(3) - \delta_D(q_i)(3) = \rho_{\mathrm{tree},i}[\delta(3, S_i) + M_{S_i}\Pi_3]$.

One-loop soliton–defect phase and KP-triangle closure (Phase 2g, `oneloop_defect_phase.pdf`). (c)

- Phonon scattering on N-soliton + CZ defect backgrounds is reflectionless ($R(q) = 0$) with factorised transmission amplitude $T_{\mathrm{tot}}(q) = \big(\prod_{j=1}^{N} A(j, \mathrm{ph}(q))\big) z_{\mathrm{ph}}(q; q_D, \eta)$.
- Setting $q_D = -(\cos\gamma)/\sqrt{3}$, the defect phase-shift derivative is a *2-root* $(-\eta, +\eta)$ *dipole* $\theta'_D(q; s, \eta) = \mathrm{Re}\sum_{s'=1}^{2} c_{D,s'}\Theta'(q; \psi_{s'})$ at $\psi_{1,2} = \pm 2\pi/3 - \gamma$, whose difference across (p_a, p_b) reproduces the 3-root $(1, 1, -2)$ soliton kernel $\theta'_S(q; \alpha, \epsilon)$.
- The defect sewing determinant factors into two quadratics **(c)**. The printed prefactor misses a factor of sixteen, and the quoted growing-mode pair misses an admissible-root branch change. Independent checks away from zeros and at $s = 1.9$ expose these errors (`audit_frames_defects.pdf`); the old frequency formula is not retained as a complete spectrum.
- With the existing zero-endpoint branch, the raw and antisymmetrised soliton–defect continuum integrals coincide, and chord differences obey formal fusion identities **(c)**. Physical quantum absorption and fusion require more: the defect carries a localized momentum operator, its boundary renormalisation is not supplied by the bulk tadpole alone, and unstable intermediate vacua cannot simply be treated as asymptotic states **(d)**(`audit_charges_action.pdf`, `audit_frames_defects.pdf`). These identities do not select the absolute scattering phase or exclude a candidate through its exchange constant alone.

Universal KP-Coboundary Master Function (Phase 2h, `charge_frame_master.pdf`). (c) For *any* odd ratio kernel $K(z) = -K(-z)$, the 2×2 dipole–dipole **two-vertex master function** on the KP root space (`scripts/kp_master_and_toda_edge.py`),

$$\mathcal{F}_K(p, q) := \sum_{s_1=1}^{2}\sum_{s_2=1}^{2}(-1)^{s_1+s_2}\,\mathrm{Re}\,K\left(\tan\frac{\psi_{s_1}(p)}{2}\cot\frac{\psi_{s_2}(q)}{2}\right), \tag{4}$$

defines the abstract soliton–defect functional as its first difference $\mathcal{I}_{SD}[K](\alpha, \epsilon; q_D, \eta) = \eta(\mathcal{F}_K(p_b, q_D) - \mathcal{F}_K(p_a, q_D))$ and the two-soliton functional as its 4-vertex second difference

$$\boxed{\mathcal{I}_{SS}[K](1,2) = \mathcal{F}_K(p_{1,b}, p_{2,b}) - \mathcal{F}_K(p_{1,a}, p_{2,b}) - \mathcal{F}_K(p_{1,b}, p_{2,a}) + \mathcal{F}_K(p_{1,a}, p_{2,a})} \tag{5}$$

guaranteeing algebraic chord closure for these defined functionals **(c)**. This does not prove that they equal the physical raw integrals in every orientation. Independent quadrature finds orientation-dependent multiples of π in the Phase 2h identification, and oddness of K alone does not imply exchange antisymmetry (`independent_phase2h_audit.pdf`, **(c)**). The decomposition $K_L = K_{\chi_2} + K_{\ln}$ remains useful algebraically; its proposed physical cancellation against the quantum momentum conversion and its all-loop extension remain unestablished **(d)**.

7 Consistency checks across candidates and orders

Claim	Audited status and evidence	Label
Classical phase	Symplectic scattering and Darboux phase checked by `symplectic_scattering.py` and `semiclassical_phase.py`	**(c)**
Continuum density	Direct potential-mode quadrature confirms the one-soliton density identity; `audit_determinant_foundations.py`	**(c)**
Collective volume	Tail cross terms cancel from the finite symplectic volume; verified through collision. The projector does not preserve fixed potential endpoints; `audit_projected_measure.py`	**(c)**
Raw phase exchange	Raw integrals are not antisymmetric and do not vanish on the diagonal; the physical exchange constant remains open	**(c)**/**(d)**
Collective charges	Regulated mode-sum L_1 and its derivative agree with the proposed formula; full field-theory identification remains conditional; `audit_charges_action.py`	**(c)**/**(d)**
Defect spectrum	Prefactor and growing-branch errors reproduced; `audit_frames_defects.py`	**(c)**
KP closure	Abstract chord identity passes; physical fusion requires normalised states and boundary renormalisation	**(c)**/**(d)**
Toda endpoint test	Exact dressed edge momentum and phase reciprocity verified; the classical frame leaves a nonseparable finite-$\hbar_T$ remainder; `audit_toda_dictionary.py`	**(c)**
Toda matching	Classical scaling passes; quantum dictionary and proposed limiting residual remain open. Saved fits do not resolve the subleading sign; `audit_phase2h_foundations.py`	**(c)**/**(d)**
Full one-loop S-matrix	Projected determinant, completeness, phase normalisation, and fixed-charge comparison are not established	**(d)**

8 Conclusions and ranked open problems

Conclusions after the independent audits.

1. *The classical symplectic phase is supported by the existing checks* **(c)**. The continuum one-loop integrals are explicit, but a fully normalised physical one-loop S-matrix and its quantum Toda reconciliation remain open **(d)**. Thus "completely solved and reconciled" is withdrawn.

2. *Candidate comparisons must specify their assumptions.* Numerical Toda discrepancies are checks within the implemented dictionary **(c)**; continuum dilogarithms and formal defect fusion do not yet provide independent, unconditional physical exclusions **(d)**.

3. *KP chord differences provide a useful algebraic representation* **(c)**. Their extension to every physical soliton observable or to an all-loop kernel is a conjecture **(d)**.

Ranked next steps. The priorities below are research judgments **(d)**.

1. **Derive the common quantum momentum and frame dictionary.** Track dressed Toda momentum, the quantum edge reference, collective soliton charges, and the frame contribution at fixed physical observables, using the new endpoint reciprocity identity as a check. Continue the direct Toda expansion independently, including logarithmic terms and separate numerical errors; do not use the old fitted $c_1(r)$ values as established two-loop targets.

2. **Construct the projected one-loop determinant.** Build on the verified finite collective symplectic density. Specify the spatial domain and admissible endpoint variations, then derive the field and constraint measures, projection, time-boundary determinant, continuum completeness, and phase normalisation together.

3. **If pursuing quantum defect fusion, derive boundary charges and renormalisation first.** Retain admissible spectral branches and distinguish stable external states from unstable intermediate configurations.

4. **Audit the uniformity of the Toda expansion when available.** Check logarithmic remainders, nearly equal amplitudes, and the vanishing smaller-soliton limit. Establish which limits commute before identifying coefficients with continuum loop orders.

5. **Revisit all-loop kernels and ring quantisation after these tests.** Quantum-dilogarithm guesses and twisted Bethe equations remain possible research directions; they should not be constrained by unproved phase constants or unresolved fitted coefficients. Non-ultralocal quantum inverse scattering remains a separate longer-term route.

A Phase 2 Write-ups and Verification Scripts

This revision changes the evidence status of the quantum synthesis; it does not introduce new external references or claim to have reread the inherited bibliography. Literature-reading scope is recorded in the independent notes. All write-ups are in `phase2/*.pdf` and all scripts are in `phase2/scripts/*.py` (plus `corrigan-zambon/defect_*.py`); each script ends with `ALL CHECKS PASSED`.

- **Phase 2a** (`report.pdf`): `scripts/symplectic_scattering.py`, `scripts/semiclassical_phase.py`, `scripts/fit_R.py`, `scripts/consistency_checks.py`, `scripts/routes_BC_checks.py`.
- **Phase 2b** (`toda_route.pdf`): `scripts/toda_bethe_classical.py`, `scripts/toda_bethe_quantum.py`, `scripts/toda_ba_vs_label.py`, `scripts/toda_scaling_fast.py`, `scripts/toda_quantum_correction.py`.
- **Phase 2c** (`smatrix_ansatz.pdf`, `defect_semiclassics.pdf`): `scripts/smatrix_ansatz.py`, `scripts/smatrix_boot.py`, `scripts/defect_smatrix_good_continuation.py`, `corrigan-zambon/defect_*.py`.
- **Phase 2d** (`toda_edge.pdf`, `oneloop.pdf`): `scripts/toda_edge_ba.py`, `scripts/toda_edge_twoterm.py`, `scripts/oneloop_step0_wellposedness.py`, `scripts/oneloop_phase_shifts.py`, `scripts/oneloop_soliton_energy.py`, `scripts/oneloop_breather.py`.
- **Phase 2e** (`oneloop_phase.pdf`): `scripts/oneloop_good_phonon_scattering.py`, `scripts/oneloop_good_twosoliton_phase.py`, `scripts/oneloop_good_limits_cusp_resonance.py`.

- **Phase 2f (`matching.pdf`):** `scripts/matching_conventions.py`, `scripts/matching_toda.py`, `scripts/matching_analysis.py`.
- **Phase 2g (`oneloop_defect_phase.pdf`):** `scripts/oneloop_defect_scattering_and_stability.py`, `scripts/oneloop_defect_phase_and_fusion.py`.
- **Phase 2h (`charge_frame_master.pdf`):** `scripts/charge_and_frame_audit.py`, `scripts/kp_master_and_toda_edge.py`.
- **Independent audits:** `independent_phase2h_audit.pdf`, `audit_charges_action.pdf`, `audit_determinant_foundations.pdf`, and `audit_frames_defects.pdf`, with the corresponding `audit_*.py` scripts listed in the consistency table.
- **Follow-up audits:** `audit_projected_measure.pdf` and `audit_toda_dictionary.pdf`, checked by the corresponding named scripts and saved JSON results.

Failures along the way (Phase 2a; later phases record their own failures in each write-up).

- The direct symbolic simplification of the symplectic identity timed out, so we switched to 40-digit numerics.
- The good-equation phase first used the wrong sign of ℓ, and the test failed until it was corrected.
- A tolerance set below double precision caused spurious failures.
- The KdV-limit comparison with Bonnemain–Doyon *does not* agree. This is recorded as a finding, not a passed test.

Failures recorded by the independent quantum audits.

- Initial PDF builds had overfull filename lines and a temporary text-group scoping error; line breaking and group closure were repaired before the final rendered-page inspection.
- During this report revision, the Phase 2h audit's source-snapshot guard failed because the report was edited while the script was running. The guard was retained; verification was repeated after editing stopped.
- The classical momentum derivatives in this report omitted the orientation factor ϵ and therefore applied only to right-movers. Both signed identities are now checked in `audit_projected_measure.py`.
- Raw-phase antisymmetry, zero diagonal, and orientation-independent master identification failed independent continuum checks.
- The displayed defect determinant prefactor and the complete growing-mode pair failed checks away from zeros and between the old sampled parameters.
- The proposed Toda residual was used to define part of its purported derivation; saved extrapolation alternatives changed the subleading sign. The claimed edge-reference explanation of a ratio-dependent slope also failed.
- Potential amplitude modes have nonlocal tails, invalidating the stated localization-only cancellation argument. Equal endpoint frequencies do not by themselves establish localized determinant cancellation.

- Profile, weighted-sum, and derivative-sign errors in intermediate charge formulas were found although the final conditional charge derivative passed. Boundary counterterms and localized defect momentum were not accounted for in the claimed physical absorption argument.

Development failures, repairs, precision, and the scope of each successful `ALL CHECKS PASSED` run are recorded in the audit notes. A successful algebraic script does not certify the unresolved physical identifications.

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